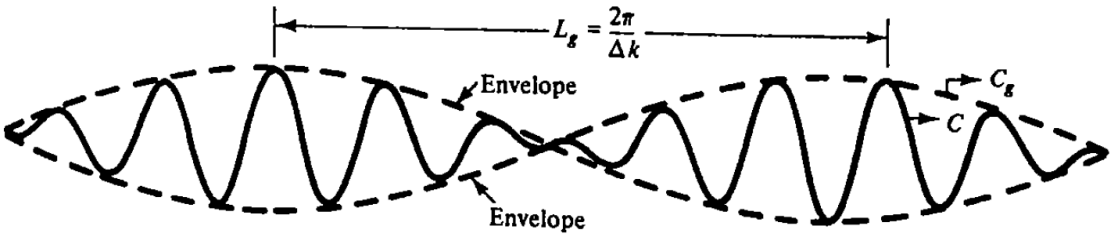
-Teoría de ondas-

Ejercicio 5

- Durante un ensayo un ensayo de una plataforma eólica marina, sometemos a la estructura a un oleaje bicromático o grupo de ondas de las siguientes características:
    - H_1=2 m / T_1 = 5.7 s
    - H_2 = 2 m / T_2 = 5.4 s
    - h = 102 m
- Calcular las características de la onda resultante:
    - Onda portadora: Periodo, número de onda y celeridad
    - Onda modulante: Periodo
    - Grupo: Periodo, Celeridad



Onda portadora

$$\omega_1 = \omega - \frac{\Delta\omega}{2} \;;\; \omega_2 = \omega + \frac{\Delta\omega}{2} \rightarrow \omega = 1{,}1329\ (T = 5{,}54s)$$

$$\omega^2 = gk\tanh kh \rightarrow 1.1329^2 = k \cdot 102 \cdot \tanh k \cdot 202$$

$$k = 0.13112\ m^{-1}$$

$$c = \frac{\omega}{k} = 8.64\ m/s$$

Onda modulante (envolvente):

$$k_1 = k - \frac{\Delta k}{2} \;;\; k_2 = k + \frac{\Delta k}{2} \rightarrow k_1 = 0{,}1238\ (L = 50{,}73m)\ ;\ k_2 = 0{,}138\ (L = 45{,}53m) \rightarrow \Delta k = 0{,}0141$$

$$\Delta\omega = 0{,}0612;$$

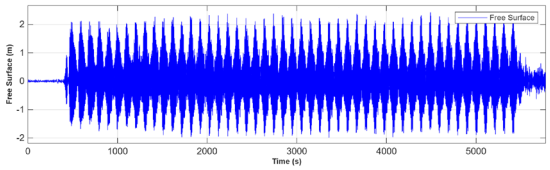
$$\eta = H\cos(kx - \omega t)\cos\left[\frac{1}{2}\Delta k\left(x - \frac{\Delta\omega}{\Delta k}t\right)\right]$$

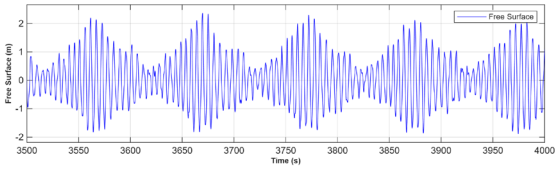
$$T_{modulante} = \frac{4\pi}{\Delta\omega} = \frac{4\pi}{0{,}0612} = 205{,}3\ \ s$$

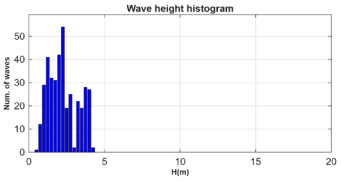
Grupo:

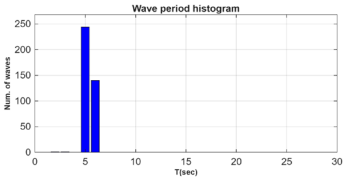
$$C_g = \frac{\Delta\omega}{\Delta k} = \frac{0.0612}{0.0141} = 4.34m/s$$

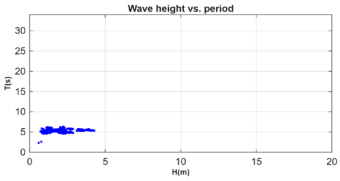
$$T_g = \frac{2\pi}{\Delta\omega} = 102.65$$

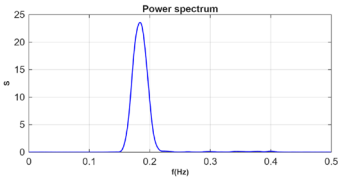












Free Surface Statistics:

Wave Height:

- H_{mean}=2.275 m
- H_{std}=0.967 m
- Hs = 3.453 m
- H_{max} = 4.264 m

Mean Period :

- T_{mean}=5.382 s

CCOB
CANTABRIA COASTAL AND OCEAN BASIN
IH cantabria
INSTITUTO DE HIDRÁULICA AMBIENTAL
UNIVERSIDAD DE CANTABRIA
Offshore Engineering and Ocean Energy Group